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Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-P03

Past Paper 3

Time: 25 minutes | Paper Reference **PMAT2/N03**

Mathematics

Level 2

Section A (Non-Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
16 /16

My signature confirms that I will not discuss the content of the test with anyone.

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Instructions

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- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used**.
- Take the value of π to be 3.14

Information

- The total mark for this section is 16.
- The marks for **each** question are shown in brackets
  – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1 Kate is making some pastry.
She will use 20 ounces of flour.

Kate has a recipe that uses 8 ounces of flour with 3 ounces of butter.

Work out how many ounces of butter Kate needs to use.

(3)

Tutor Strategy: Proportion & Scale Factor

- *Find the scale factor by dividing total flour needed (20 oz) by recipe flour (8 oz).*
- *Multiply recipe butter (3 oz) by this scale factor to find the exact butter required.*

Method 1: Unitary Scale Factor

Scale Factor = $\frac{20}{8}$ ***= 2.5*** *(needs 2.5 times the recipe)* **[M1]**

Butter needed = 3 × 2.5 **[M1]**

3 × 2 = 6, 3 × 0.5 = 1.5

6 + 1.5 = 7.5 ounces **[A1]**

Method 2: Ratio Chunking Check

- *8 oz flour = 3 oz butter*
- *8 oz flour = 3 oz butter*
- *4 oz flour = 1.5 oz butter (halving both sides)*
- *Total: 8 + 8 + 4 = 20 oz flour ==> 3 + 3 + 1.5 = 7.5 oz butter (matches)*

7.5 ✓ ounces

(Total for Question 1 is 3 marks)

2 (a) Work out the value of $4x + 3$ when $x = 2$

(1)

Substitute x = 2 into 4x + 3:

4x + 3 = 4(2) + 3

= 8 + 3 = 11

BIDMAS Rule

- 4x means 4 × x.
- Always multiply first before adding (+ 3).

[A1]

11

Here is a formula

$A = 5\sqrt{B}$

(b) Work out the value of A when $B = 16$

(2)

Formula: A = 5 sqrt(B)

Substitute B = 16:

A = 5 × sqrt(16)

[M1]

sqrt(16) = 4 (since 4 × 4 = 16)

[A1]

A = 5 × 4 = 20

20

(Total for Question 2 is 3 marks)

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3 (a) Work out $\frac{5}{8}$ of 320

(2)

Step 1: Divide by denominator to find 1/8 of 320:

320 ÷ 8 = 40 (Mental tip: 32 ÷ 8 = 4, so 320 ÷ 8 = 40) [M1]

Step 2: Multiply by numerator (5) to find 5/8 of 320:

40 × 5 = 200 [A1]

200 ✓

Kanya makes and sells jewellery items.
In one week she sells earrings, bracelets and necklaces in the ratio 13 : 24 : 15

(b) What fraction of the total items sold were bracelets?
Give your answer in its simplest form.

(2)

Ratio = Earrings : Bracelets : Necklaces
= 13 : 24 : 15

Total parts = 13 + 24 + 15 = 52 parts [M1]

Fraction of bracelets = 24 / 52

Divide numerator & denominator by 4:

Simplest form = 6 / 13 [A1]

$\frac{6}{13}$ ✓

(Total for Question 3 is 4 marks)

4 Paul sees this advert for a payment plan for a laptop.

Laptop payment plan

normal price £240

pay a deposit of £36 and £15 a month for 12 months

use our payment plan and save 8% of the normal price

(a) Does using the payment plan save 8% of the normal price?
You **must** show your working.

(5)

Pedagogical Plan: Compare Payment Plan vs 8% Discount

1. Calculate total payment plan cost (deposit + 12 monthly instalments).

2. Calculate 8% of £240 and compare savings to decide if plan saves >= 8%.

Step 1: Calculate total payment plan cost

Deposit = £36

12 instalments @ £15 = 12 × 15 = £180 *(10×15=150, 2×15=30)* **[M1]**

Total plan cost = £36 + £180 = £216 **[M1]**

Step 2: Calculate 8% of £240 (without a calculator)

- **10% of £240 = £240 ÷ 10 = £24.00**
- **1% of £240 = £2.40 ==> 2% = 2 × £2.40 = £4.80**
- **8% of £240 = 10% - 2% = £24.00 - £4.80 = £19.20** **[M1]**

Discounted price = £240 - £19.20 = £220.80 **[M1]**

Step 3: Compare saving on plan vs 8% target

Actual plan saving = £240 - £216 = £24.00

Plan saves £24 (10%), which is MORE than 8% (£19.20).

(See page 7 for Alternative Check and Official Answer Box)

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Question 4(a) Continuation & Alternative Verification:

Alternative Percentage Method:

- Actual plan saving = £240 - £216 = £24.00
- Percentage saved = (£24 ÷ £240) × 100 = 10%
- Advertised target saving = 8%
- 10% saving > 8% saving ==> The payment plan saves 10%, NOT 8%.

Conclusion for Q4(a):

Does the payment plan save 8%? No – it saves 10% (£24 ≠ £19.20).

Pearson Edexcel Marking Principle:

- For decision questions, mark schemes award A1 for the correct conclusion ('No') supported by valid mathematical evidence (£216 vs £220.80, or 10% vs 8%).
- Always write 'No' clearly in the answer box and state both comparison values.

No ✓ **[A1]**

☑ (b) Use estimation to check your calculation for the cost of the payment plan.

(1)

Estimation check for payment plan cost:

- **Round deposit £36 ≈ £40; 12 × £15 = £180 (or 10 × £15 = £150)**
- **Estimated total = £40 + £180 = £220** **[B1]**

£220 is close to the exact calculated cost of £216. ✓

(Total for Question 4 is 6 marks)

TOTAL FOR SECTION A IS 16 MARKS

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Pearson Edexcel Functional Skills

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MFS-UK	L2-P03

Past Paper 3

Time: 1 hour 30 minutes | Paper Reference **PMAT2/C03**

Mathematics

Level 2

Section B (Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
48 /48

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- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used**.
- If your calculator does not have a π button take the value of π to be 3.14

Information

- The total mark for this section is 48.
- The total mark for this paper is 64.
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Advice

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- Check your answers if you have time at the end.

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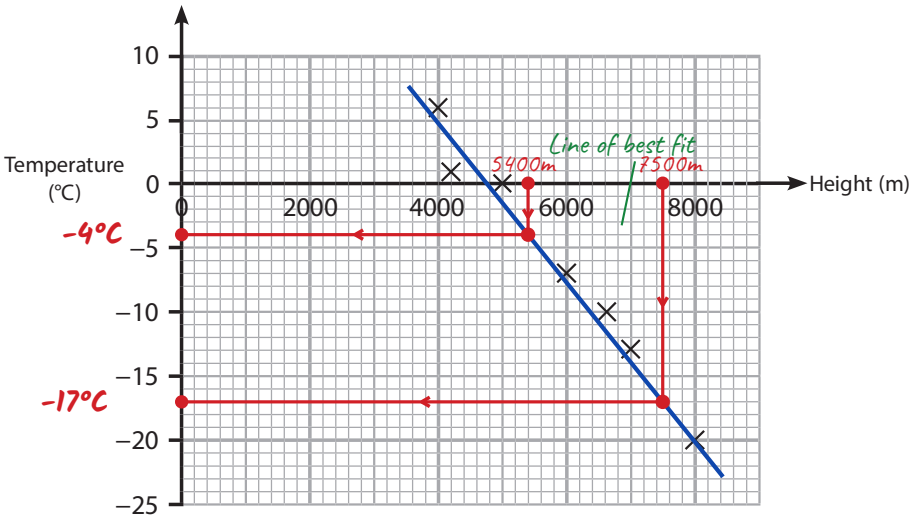
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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 The scatter diagram shows the temperature at different heights on Mount Everest for one afternoon in May.



(a) Describe the relationship shown in this scatter diagram.

(1)

Negative correlation (as height increases, temperature decreases) [A1]

There are camps on Mount Everest.

Base camp A is at a height of 5400 m.
Camp 3 is at a height of 7500 m.

(b) Use the scatter diagram to work out the difference in temperatures between the two camps.

(2)

• Base camp A (5400 m) = -4 °C, Camp 3 (7500 m) = -17 °C [M1]

Difference = -4 - (-17) = -4 + 17 = 13 °C (accept 12 to 14) [A1]

13 °C ✓

(Total for Question 1 is 3 marks)

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2 Jenni is an editor for a newspaper.
She checks information in an article.

The article states, 'In 2019 there were 300 people per square kilometre in the UK.'

Jenny knows that in 2019

- the area of the UK was 242 500 square kilometres
- the population of the UK was 66.9 million.

(a) Is the statement in the article correct?
Show why you think this.

(2)

Population = 66.9 million = 66,900,000

Population density = Total Population ÷ Land Area

= 66,900,000 ÷ 242,500 = 275.876... people / km² **[M1]**

Compare with statement (300 people / km²):

• Actual density is 275.88 people/km², which is LESS than 300.

• Or: 300 × 242,500 = 72,750,000 (UK would need 72.75m people).

No ✓ **[A1]**

☑ (b) Use a reverse calculation to show a check of your answer.

(1)

Reverse Calculation Check:

275.876... × 242,500 = 66,900,000 **[B1]**

= 66.9 million people (matches given population) ✓

(Total for Question 2 is 3 marks)

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3 Wesley is a salesman.

He wants to work out the total cost of the petrol he will use for travel next week.

Wesley knows

- he will drive a total of 520 miles
- his car uses 1 gallon of petrol per 28 miles
- petrol costs 128.4 pence per litre
- 1 gallon = 4.55 litres.

Wesley thinks the petrol he will use for travel next week will cost more than £100

Is Wesley correct?
Show why you think this.

(4)

Multistep Travel Cost Conversion

1. Miles ÷ mpg = Gallons of petrol needed.
2. Gallons × 4.55 = Litres of petrol needed.
3. Litres × pence per litre = Total pence ÷ 100 = Total pounds (£).

Step 1: Calculate gallons of petrol needed

Gallons = 520 ÷ 28 = 18.5714... gallons **[M1]**

Step 2: Convert gallons to litres (1 gal = 4.55 L)

Litres = 18.5714... × 4.55 = 84.5 litres **[M1]**

Step 3: Calculate total cost in pence & pounds

Cost in pence = 84.5 × 128.4p = 10,849.8 pence

Cost in pounds = 10,849.8 ÷ 100 = £108.498 ≈ £108.50 **[M1]**

Step 4: Compare with Wesley's estimate (£100):

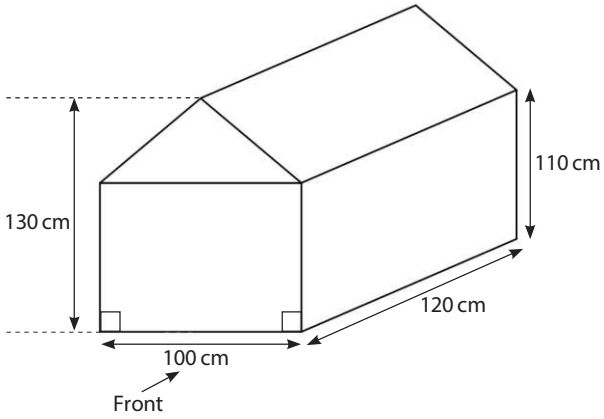
£108.50 is greater than £100. Wesley is correct.

Yes ✓ **[A1]**

(Total for Question 3 is 4 marks)

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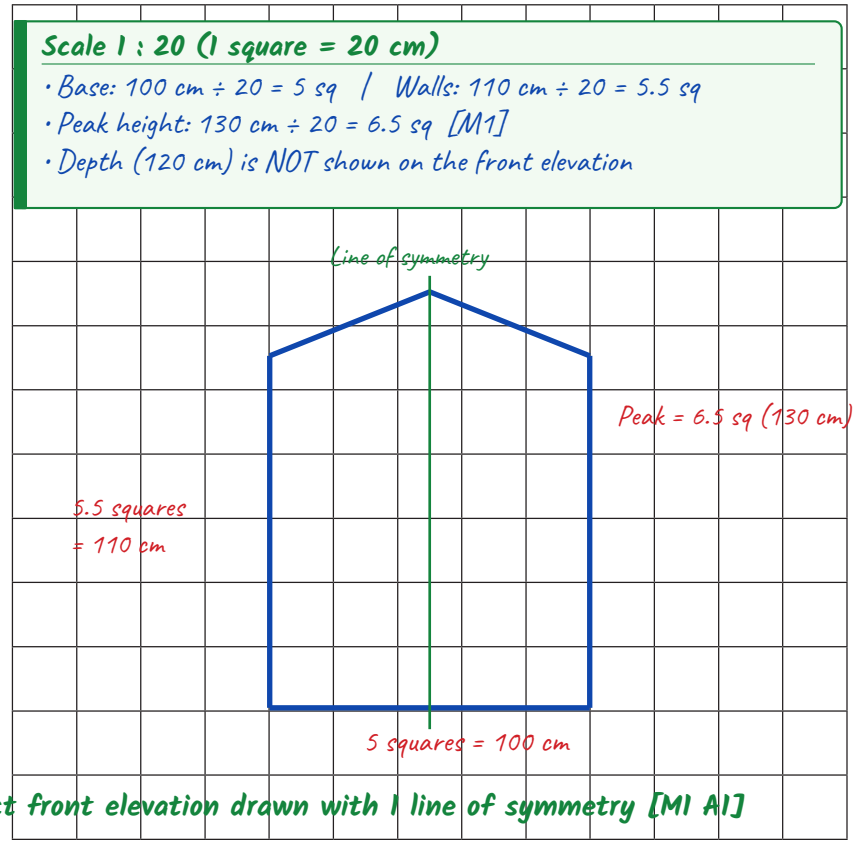
4 Anwar makes a playhouse for his children.



The front elevation of the playhouse has one line of symmetry.

Draw the front elevation of the playhouse on the grid.
Use the scale 1 : 20

(3)



(Total for Question 4 is 3 marks)

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5 In October Mr Barker gave a group of 250 students a history test.

These are the results.

Number of marks	Frequency	Midpoint (x)	f x x
1 to 5	20	3	20 x 3 = 60
6 to 10	50	8	50 x 8 = 400
11 to 15	120	13	120 x 13 = 1560
16 to 20	60	18	60 x 18 = 1080
Total	Total f = 250		Total = 3100 [M1]

Mr Barker estimates the mean mark to be 12

(a) Is Mr Barker correct?
Show why you think this.

(3)

Estimated Mean mark = Total (f x x) ÷ Total Frequency

= 3100 ÷ 250 = 12.4 [M1]

Compare with Mr Barker's estimate of 12:

- The exact estimated mean is 12.4, which is not 12.
- (Note: Mark scheme condones 'Yes' if learner rounds 12.4 ≈ 12).

No ✓ [A1]

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In the October test the lowest mark was 2 and the highest mark was 17

In November the same students took a similar test.
The lowest mark was 5 and the highest mark was 18

Mr Barker wants to write a statement comparing the spread of marks in the two tests.

(b) Write a statement to compare the spread of marks in the two tests.
You **must** show calculations to support your statement.

(2)

Measure of Spread: The Range

- *Range = Highest value - Lowest value.*
- *A smaller range means data is less spread out, more consistent, and more closely grouped.*

Step 1: Calculate the range for each test

October test range = Highest - Lowest = 17 - 2 = 15

November test range = Highest - Lowest = 18 - 5 = 13 **[M1]**

Step 2: Comparison Statement

Statement to compare spread of marks:

The range of marks in November (13) is smaller than the range in October (15).

This means the marks in November were less spread out and more consistent than the marks in October.

[A1]

(Total for Question 5 is 5 marks)

6 Mia is the manager of an ice cream shop.
In August she sold 960 ice lollies.
In September she sold 810 ice lollies.

Mia thinks she sold 18% fewer ice lollies in September than in August.

Did Mia sell 18% fewer ice lollies in September than in August?

(3)

Percentage Decrease Formula

- *Decrease = Original value (August) - New value (September).*
- *Percentage Decrease = (Decrease ÷ Original value) × 100.*

Step 1: Calculate actual decrease in ice lollies sold

Decrease = 960 − 810 = 150 ice lollies **[M1]**

Step 2: Calculate percentage decrease

Percentage decrease = (150 ÷ 960) × 100 = 15.625% **[M1]**

Alternative Check (18% of August Sales):

- *18% of 960 = 0.18 × 960 = 172.8 ice lollies.*
- *To sell 18% fewer, she would need to sell 172.8 fewer lollies (960 - 172.8 = 787.2).*
- *Since she actually sold 810 (only 150 fewer), the decrease is less than 18%.*

Comparison & Conclusion:

15.625% is LESS than 18%. Mia sold 15.625% fewer (150 ≠ 172.8).

No ✓ **[A1]**

(Total for Question 6 is 3 marks)

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7 Alex has two sets of four cards.
He writes a number on each card.

Alex picks one card from each set and multiplies the numbers to get a score.

The table shows some of the scores.

Card 1

Card 2 ×	**–5**	**7**	**–9**	**11**
2	–10	14	-18	22
4	-20	28	–36	44
–6	30	–42	54	-66
8	–40	56	-72	88

[M1 A1: 2 marks]

(a) Complete the table.

(2)

Alex says,

'The probability that the score is negative is 0.5, which means there is a 5% chance that the score is negative.'

(b) Alex is incorrect.
Explain why.

(1)

0.5 is equal to 50%, not 5%. (5% is written as 0.05). [B1]

To convert decimal to percentage: 0.5 × 100 = 50%.

Alex picks one card from each set.

(c) What is the probability that the score is 14?

(1)

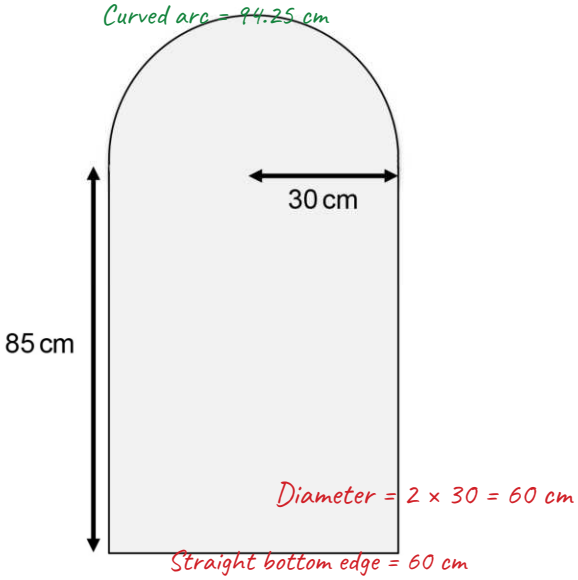
- Total outcomes in table = 4 × 4 = 16
- Score of 14: only 2 × 7 = 14 (1 outcome)

1 / 16 ✓ [A1]

(Total for Question 7 is 4 marks)

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8 Abi has a mirror in the shape of a rectangle with a semicircle on one of the shorter edges.
The radius of the semicircle is 30 cm.
The longer edge of the rectangle is 85 cm.



Abi buys a 500 cm length of mirror trim to go around all the edges of the mirror.

She thinks she will use more than $\frac{2}{3}$ of the length of the mirror trim.

Is Abi correct?
Show why you think this.

(6)

Perimeter of Compound Shape: Step-by-Step

Step 1: Identify all outer edges of the mirror:
- *2 longer straight sides: 85 cm each.*
- *1 shorter straight side (opposite semicircle): 2 × 30 = 60 cm.*
- *1 curved semicircle edge: half the circumference of a full circle.*

Step 2: Curved semicircle edge length:
- *Full circumference = 2 × pi × r = 2 × 3.142 × 30 = 188.5 cm*
- *Semicircle curved edge = 188.5 ÷ 2 = 94.25 cm (or pi × r = 30 pi ≈ 94.25 cm) [M1]*

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Step 3: Calculate total perimeter of the mirror

Total Perimeter = 85 + 85 + 60 + 94.25

= 230 + 94.25 = 324.25 cm [M1 M1]

(or 324.2 cm using pi = 3.14)

Step 4: Calculate 2/3 of the 500 cm mirror trim

2/3 of 500 cm = (500 ÷ 3) × 2

= 166.666... × 2 = 333.33 cm [M1]

Alternative Check (Fraction of trim used):

324.25 ÷ 500 = 0.6485 (64.85%)

2/3 = 0.6667 (66.67%) ==> 64.85% is LESS than 2/3.

Step 5: Compare and conclude

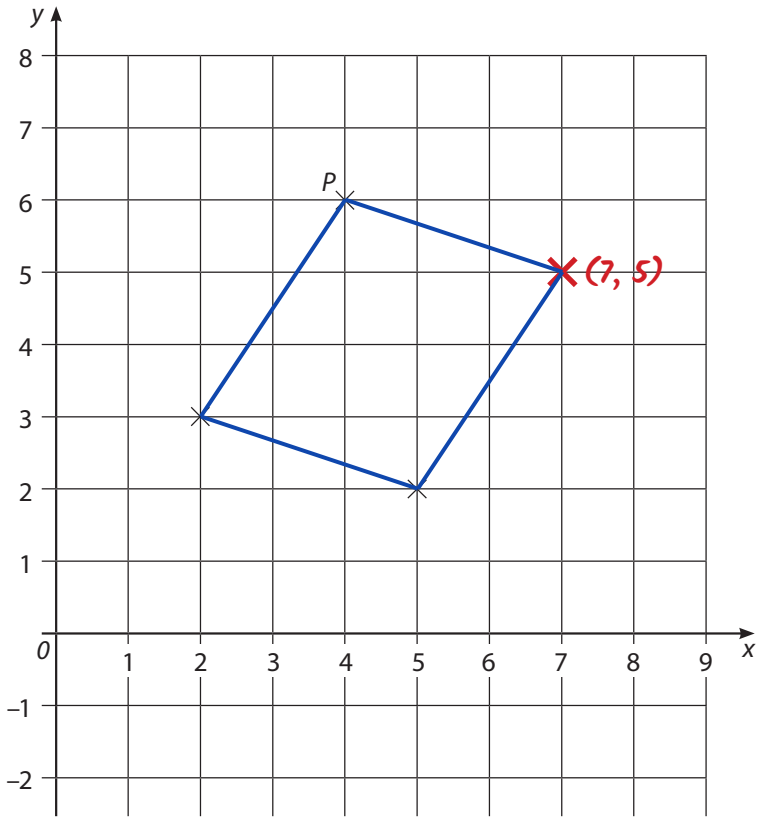
324.25 cm < 333.33 cm (uses less than 2/3 of the trim)

No ✓ [A1]

(Total for Question 8 is 6 marks)

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9 Karl plots 3 points on a grid.



(a) Write down the coordinates of the point *P*.

[A1]

(1)

(4 , 6)

Karl plots another point.
He joins the four points to make a parallelogram.

(b) Write down a possible set of coordinates for the fourth point.
Plot the point on the grid.

[M1 A1]

(2)

Shift (2, 3) to P(4, 6): +2 across, +3 up
(5 + 2, 2 + 3) = (7, 5) (or (1, 7) or (3, -1))

(7 , 5)

(Total for Question 9 is 3 marks)

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10 Andy and his brother set up eco cabins.
It takes the 2 of them 6 days to set up one eco cabin.

A friend starts working with them.

How many days will 3 people take to set up one eco cabin?

(3)

Inverse Proportion Strategy

- *More workers mean fewer days to finish the job.*
- *Total person-days required is constant: People × Days = Total work.*
- *When the team grows to 3 people, divide total person-days by 3.*

Step 1: Calculate total work required in person-days

Total work = 2 people × 6 days = 12 person-days **[M1]**

Step 2: Calculate days taken by 3 people

Days taken = 12 person-days ÷ 3 people

= 4 days **[M1 A1]**

Verification check:

3 people × 4 days = 12 person-days (constant work verified) ✓

4 ✓ days

(Total for Question 10 is 3 marks)

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11 Gavin is a car salesman.

The table shows the number of cars he sold in one week.

Mon	Tue	Wed	Thu	Fri	Sat
2	0	1	4	1	2

The cost of each car is £20 950

For every car Gavin sells he earns

- 1.25% of the cost of the car in commission
- a bonus of £50

Work out the median amount Gavin earned per day for this week.

(5)

Two-Stage Pedagogical Method

Method A: Find median number of cars sold per day, then calculate earnings.
Method B: Calculate daily earnings for all 6 days, sort in order, and take the middle value.
Both methods yield the exact same median daily earnings.

Step 1: Order daily car sales to find the median cars sold

Daily sales (ordered): 0, 1, 1, 2, 2, 4

Middle values (3rd and 4th) are 1 and 2

Median cars sold per day = (1 + 2) ÷ 2 = 1.5 cars **[M1]**

Step 2: Calculate commission and bonus per car

- **Commission = 0.0125 × £20,950 = £261.875** **[M1]**
- **Bonus = £50.00**
- **Total earnings per car = £261.875 + £50 = £311.875** **[M1]**

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Step 3: Calculate median amount earned per day

Median daily earnings = Median cars × Earnings per car

= 1.5 × £311.875 = £467.8125 ≈ £467.81 [M1 A1]

Method B: Day-by-Day Earnings Verification

Calculate daily earnings for each day:

- Tue (0 cars): £0.00
- Wed (1 car): 1 × £311.875 = £311.875 (£311.88)
- Fri (1 car): 1 × £311.875 = £311.875 (£311.88)
- Mon (2 cars): 2 × £311.875 = £623.75
- Sat (2 cars): 2 × £311.875 = £623.75
- Thu (4 cars): 4 × £311.875 = £1,247.50

Ordered list: £0, £311.88, £311.88, £623.75, £623.75, £1247.50

Median = (£311.875 + £623.75) ÷ 2 = £935.625 ÷ 2 = £467.8125 ≈ £467.81

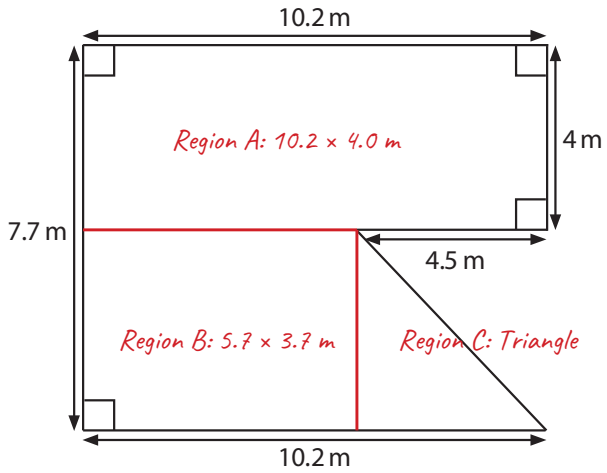
£ 467.81

(Total for Question 11 is 5 marks)

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12 Maninder is the manager of a restaurant.
She wants to cover the floor of the restaurant with tiles.

This is a plan of the floor of the restaurant.



Maninder will buy the tiles in packs.
Each pack covers an area of 0.945 m^2
One pack of tiles costs £16.15

Tiles can be cut and joined.

Work out the total cost of the packs of tiles Maninder will buy.

(6)

Step 1: Find missing dimensions of the compound shape

- Horizontal bottom-left width = 10.2 − 4.5 = 5.7 m
- Vertical bottom-left height = 7.7 − 4.0 = 3.7 m **[M1]**

Step 2: Calculate areas of the 3 regions

- Area A (top rectangle) = 10.2 × 4.0 = 40.8 m^2
- Area B (bottom-left rectangle) = 5.7 × 3.7 = 21.09 m^2
- Area C (bottom-right triangle) = 1/2 × 4.5 × 3.7 = 8.325 m^2 **[M1]**

Total Floor Area = 40.8 + 21.09 + 8.325 = 70.215 m^2 **[M1]**

(Alternative subtraction check: 10.2 × 7.7 = 78.54; 78.54 − 8.325 = 70.215 m^2)

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Packs & Rounding Golden Rule

- Tiles come in whole packs. If division gives a decimal, always ROUND UP.
- Rounding down would leave parts of the floor uncovered!

Step 3: Calculate number of packs needed

Each pack covers 0.945 m²

Packs needed = 70.215 ÷ 0.945

= 74.3015... packs [M1]

Round UP to nearest whole pack:

Packs to buy = 75 packs [M1]

Step 4: Calculate total cost of tiles

Cost = Number of packs × Price per pack

= 75 × £16.15 = £1,211.25 [M1 A1]

Reverse check:

£1,211.25 ÷ £16.15 = 75 packs ✓

£ 1,211.25 ✓

(Total for Question 12 is 6 marks)

TOTAL FOR SECTION B IS 48 MARKS
TOTAL FOR PAPER IS 64 MARKS

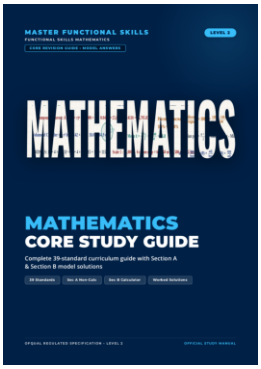
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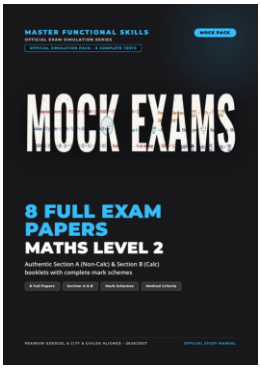


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